SentinelQ

An automated signal engine for portfolio monitoring — executive brief on methodology, reliability and production readiness

SentinelQ reads the public news, disclosures and corporate-action history of every stock in a portfolio and turns them into three independent, dated, source-linked scores: how the market narrative is trending, whether anything adverse has happened in governance, and how much risk is sitting in recent corporate actions. It is built to be checked, not trusted blindly — every number traces back to a specific article and a fixed, published rule, and the same inputs always produce the same output. This brief summarises what it does, why its output can be relied on, what has been built and tested so far, and what remains before it is a production system of record rather than a prototype.

What it produces

For every stock in a portfolio, SentinelQ scores three separate dimensions, each on its own fixed scale:

Dimension	Scale	What it captures
Sentiment	-2 to +2	How the past year of reported news reads for the company — results, guidance, orders, analyst views — weighted so recent news counts more than old news. Price moves never feed this score; only reported, dated events do.
Governance	0 to 100	Starts clean and loses fixed points only for specific, verified adverse events (a regulator's order, an auditor's exit, an unplanned senior departure, a promoter pledge being called). Routine events — a scheduled retirement, a resolved inquiry — are reviewed and explicitly not penalised.
Corporate Actions	0 to 10	How much risk is concentrated in a stock's recent capital, ownership or structural activity — a dilutive raise, a change of control, a promoter pledge — combined so one serious event is never diluted by several minor ones.

A fourth, read-only layer — dividends, buybacks, splits and similar mechanical actions — is listed for context but deliberately kept out of all three scores: these do not change what a company is worth, and including them would distort the signal rather than inform it.

How it works

The system follows a fixed, six-stage pipeline. Only one stage uses AI, and its role is deliberately narrow: it reads one news article at a time and assigns it a label from a closed, pre-defined list — an event type, a sentiment value on a fixed scale, and whether it concerns governance. It never computes a score, never sees the scoring rules, and never aggregates anything across articles. Every other stage — retrieving the news, filtering out noise, turning labels into scores, writing the report — is ordinary, deterministic software with no AI involved and no randomness.

The labelling step recognises 45 distinct event types spanning results, governance, and the full range of formal corporate actions (capital raises, mergers, buybacks, pledges, insolvency and more), each mapped to a documented severity and treatment.

Stage	What happens
1. Input	Resolve the exact company identity (name, ticker, sector) so every later search is unambiguous.

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2. Collect	Retrieve news, results coverage and corporate-action history for the lookback period; discard templated and off-topic items before anything is read in depth; merge duplicate coverage of the same story into one event so repetition is never counted as repeated evidence.
3. Label	The one AI step: read each distinct event and assign it a fixed-vocabulary label — nothing else.
4. Validate	Check every label against the expected structure; anything malformed is retried once, then dropped with a recorded reason — never silently kept or silently discarded.
5. Score	Deterministic arithmetic only. Turn labels into the three scores using fixed, published rules — no AI, no judgement calls, fully reproducible by hand from the same inputs.
6. Report	Produce the scorecard and the full supporting evidence — every score traceable to the specific, dated, linked articles behind it.

Why the output can be trusted

Deterministic scoring. The scoring rules live in one published file, separate from the AI. Given the same labelled evidence, the system produces the identical score every time — nothing is re-derived by a model at report time, and nothing drifts between runs.

Full traceability. Every score is backed by a visible list of the specific articles that drove it, each with its date, headline, individual assessment, and link. Nothing is a black-box average — an analyst can open any score and see exactly why it is what it is.

Source-strength transparency. Each piece of evidence is weighted by how many independent outlets reported it and how reliable those outlets are, on a fixed, editable scale. A claim backed by a single low-reliability source is explicitly flagged as such in the output, rather than presented with the same confidence as a well-corroborated one.

Nothing proceeds unverified. Adverse governance events pass a verification check before they can cost any points — confirming the event genuinely concerns the company itself, not a peer, a counterparty, or an industry-wide rule that happens to share a keyword. Items that fail this check are visibly logged as reviewed and excluded, not silently dropped.

Nothing silently fails. An article that cannot be read, a label that doesn't parse, a stock that cannot be resolved — every one of these produces a recorded reason in the output, never a quiet omission. If more than a small fraction of a run fails, the whole run stops rather than publishing on unreliable foundations.

What has been built and tested

The current system is a working prototype, not a paper design: it runs end to end against live, public news sources and produces the full scorecard and supporting workbook for any named list of stocks today. The deterministic scoring logic — the part of the system that turns evidence into a number, and the part that matters most for consistency and auditability — is covered by an automated test suite of 166 checks that run before any change is accepted, each one pinned to a specific, hand-verified expected result. The system has been run and reviewed against real portfolios of varying size, with its output checked line by line against the underlying articles.

What production readiness would add

The prototype demonstrates the methodology works. Moving it to a system the firm relies on day to day would mean addressing the following, in roughly the order they matter:

Area	What it would mean
Primary-source data	Today's evidence is drawn from public news coverage. Direct integration with exchange filings and regulatory disclosures would reduce dependence on how well a given event happened to be reported, and is the single highest-value addition for governance and corporate-action coverage specifically.
Human sign-off workflow	A defined point at which a person reviews and approves scores before they reach decision-makers — particularly for the lowest-scoring names and anything a verification check could not fully resolve on its own.
Scheduled, monitored operation	Running on a fixed schedule with alerting if a run fails or produces an unexpected shift, rather than being run on request.
Coverage breadth and calibration	Testing across a wider and more varied set of names than has been exercised so far, to confirm the scoring thresholds hold up across sectors and company sizes, with a feedback loop to tune them as real outcomes come in.
Access and distribution	A defined way for the people who need these scores to receive them — a shared location, a scheduled distribution, or integration into an existing review process.

Recommendation

The methodology is sound, deterministic, and fully auditable, and the prototype already produces usable output against live data. The most efficient next step is a short pilot against a defined, representative portfolio, with a named reviewer validating a sample of scores against source filings before anything from the system reaches a wider audience. That pilot would surface any remaining gaps directly, rather than theorising about them, and gives a concrete basis for deciding the investment in the production items above.

This is a summary for decision-making purposes. The full technical reference — every scoring rule, formula and worked example — is available on request.